

Compendium of Formulas, Rules, and Conditions

Chapter 5: Linear Inequalities

Authoritative Mathematical Reference Manual

1 Fundamental Definitions and Concepts

1.1 Definition of an Inequality

An algebraic or numerical statement involving real numbers or variables connected by inequality relation symbols is defined as an **inequality**.

Relational Symbols

- $<$: Strictly less than
- $>$: Strictly greater than
- $\leq$: Less than or equal to (at most)
- $\geq$: Greater than or equal to (at least)

1.2 Solution and Solution Set

- **Solution:** Any real number value assigned to the variable that converts the inequality into a true mathematical statement.
- **Solution Set:** The set containing all possible solutions of the given inequality.
- **Replacement Sets (Domains):**
 - Natural Numbers ($\mathbb{N}$): Solutions restricted to $\{1, 2, 3, \dots\}$.
 - Integers ($\mathbb{Z}$): Solutions restricted to $\{\dots, -2, -1, 0, 1, 2, \dots\}$.
 - Real Numbers ($\mathbb{R}$): Continuous intervals on the real number line.

2 Classification of Inequalities

2.1 Categorization by Structure

Category	Mathematical Form / Condition	Examples
Numerical Inequality	Involves only constant real numbers	$3 < 5, 7 > -2$
Literal Inequality	Involves one or more variables	$x < 5, y \geq 4$
Double Inequality	Combined bounded inequality relation	$3 < x \leq 7, a \leq f(x) < b$

2.2 Categorization by Relation Type

1. **Strict Inequalities:** Expressed strictly with $<$ or $>$ (no equality permitted).

$$ax + b < 0, \quad ax + b > 0, \quad ax + by < c, \quad ax + by > c$$

2. **Slack (Non-Strict) Inequalities:** Expressed with $\leq$ or $\geq$ (includes equality boundary).

$$ax + b \leq 0, \quad ax + b \geq 0, \quad ax + by \leq c, \quad ax + by \geq c$$

2.3 Categorization by Degree and Variables

- **Linear Inequality in One Variable:**

$$ax + b < 0, \quad ax + b \leq 0, \quad ax + b > 0, \quad ax + b \geq 0 \quad (a \neq 0, a, b \in \mathbb{R})$$

- **Linear Inequality in Two Variables:**

$$ax + by < c, \quad ax + by \leq c, \quad ax + by > c, \quad ax + by \geq c \quad (a \neq 0, b \neq 0, a, b, c \in \mathbb{R})$$

- **Quadratic Inequality in One Variable:**

$$ax^2 + bx + c < 0, \quad ax^2 + bx + c \leq 0 \quad (a \neq 0, a, b, c \in \mathbb{R})$$

3 Axiomatic Rules for Solving Linear Inequalities

When transforming or solving linear inequalities algebraically, the following two strict fundamental rules govern all operations:

Rule 1: Addition and Subtraction Invariance

Equal real numbers or expressions may be added to (or subtracted from) both sides of an inequality **without changing** the direction of the inequality sign:

$$A < B \iff A + C < B + C \quad \text{and} \quad A < B \iff A - C < B - C \quad (\forall C \in \mathbb{R})$$

Rule 2: Multiplication and Division Transformations

1. **Positive Multiplier/Divisor ($k > 0$):** Multiplying or dividing both sides by a positive number preserves the inequality direction:

$$A < B \iff kA < kB \quad \text{and} \quad A < B \iff \frac{A}{k} < \frac{B}{k} \quad (k > 0)$$

2. **Negative Multiplier/Divisor ($k < 0$) — SIGN REVERSAL:** Multiplying or dividing both sides by a negative number **REVERSES** the direction of the inequality sign:

$$A < B \iff kA > kB \quad (k < 0)$$

$$A \leq B \iff kA \geq kB \quad (k < 0)$$

$$A > B \iff kA < kB \quad (k < 0)$$

$$A \geq B \iff kA \leq kB \quad (k < 0)$$

4 Real Interval Notation and Representations

For real solutions ($x \in \mathbb{R}$), solutions are represented using interval notation:

Interval Type	Notation	Set-Builder Form	Boundary Condition
Open Interval	(a, b)	$\{x \in \mathbb{R} : a < x < b\}$	Excludes both a and b
Closed Interval	$[a, b]$	$\{x \in \mathbb{R} : a \leq x \leq b\}$	Includes both a and b
Left-Closed, Right-Open	$[a, b)$	$\{x \in \mathbb{R} : a \leq x < b\}$	Includes a , excludes b
Left-Open, Right-Closed	$(a, b]$	$\{x \in \mathbb{R} : a < x \leq b\}$	Excludes a , includes b
Unbounded Upper (Open)	(a, ∞)	$\{x \in \mathbb{R} : x > a\}$	Strictly greater than a
Unbounded Upper (Closed)	$[a, \infty)$	$\{x \in \mathbb{R} : x \geq a\}$	Greater than or equal to a
Unbounded Lower (Open)	$(-\infty, b)$	$\{x \in \mathbb{R} : x < b\}$	Strictly less than b
Unbounded Lower (Closed)	$(-\infty, b]$	$\{x \in \mathbb{R} : x \leq b\}$	Less than or equal to b

5 Graphical Representation Conventions on Number Line

To represent the solution set of a linear inequality in one variable on the real line:

- **Strict Endpoint Boundary ($<$ or $>$):** Marked with a **hollow circle** ($\circ$) at the endpoint to indicate that the boundary point is *excluded*.
- **Slack Endpoint Boundary ($\leq$ or $\geq$):** Marked with a **solid circle** ($\bullet$) at the endpoint to indicate that the boundary point is *included*.
- **Directional Ray:** A thick line or shaded ray is drawn along the real line in the direction satisfying the inequality:
 - Towards $+\infty$ (rightward) for $x > a$ or $x \geq a$.
 - Towards $-\infty$ (leftward) for $x < b$ or $x \leq b$.

6 System of Simultaneous Linear Inequalities

6.1 Definition and Solution Concept

A system of linear inequalities consists of two or more inequalities in the same variable(s). The solution set of the system S is the **intersection** of the solution sets of individual inequalities:

$$S = S_1 \cap S_2 \cap S_3 \cap \cdots \cap S_n$$

6.2 Double Inequalities

An inequality of the form $a \leq f(x) \leq b$ is equivalent to the simultaneous compound system:

$$a \leq f(x) \leq b \iff f(x) \geq a \quad \text{and} \quad f(x) \leq b$$

To solve $-a \leq px + q \leq b$ directly:

$$-a - q \leq px \leq b - q \implies \frac{-a - q}{p} \leq x \leq \frac{b - q}{p} \quad (\text{for } p > 0)$$

7 Key Applied Modeling Formulas and Conditions

7.1 Temperature Conversion Formula

The relation between temperature in Celsius (C) and Fahrenheit (F) is:

$$C = \frac{5}{9}(F - 32) \iff F = \frac{9}{5}C + 32$$

For a solution required to be kept between $T_1^\circ\text{C}$ and $T_2^\circ\text{C}$ ($T_1 < C < T_2$):

$$T_1 < \frac{5}{9}(F - 32) < T_2 \iff \frac{9}{5}T_1 + 32 < F < \frac{9}{5}T_2 + 32$$

7.2 Mean / Average Condition

For n test scores $x_1, x_2, \dots, x_n$, to achieve an average score of at least μ :

$$\frac{x_1 + x_2 + \dots + x_n}{n} \geq \mu \iff \sum_{i=1}^n x_i \geq n\mu$$

7.3 Consecutive Integer / Real Bounds

- Two consecutive odd (or even) positive integers: x and $x + 2$.
- Sum bound condition ($< K$): $x + (x + 2) < K \implies 2x + 2 < K$.
- Individual magnitude condition ($> M$): $x > M$.

7.4 Geometric Triangle Inequality Conditions

For a triangle with side lengths a, b, c :

$$a + b > c, \quad b + c > a, \quad a + c > b$$

Perimeter $P = a + b + c$. If $P \geq P_{\min}$, then $a + b + c \geq P_{\min}$.